

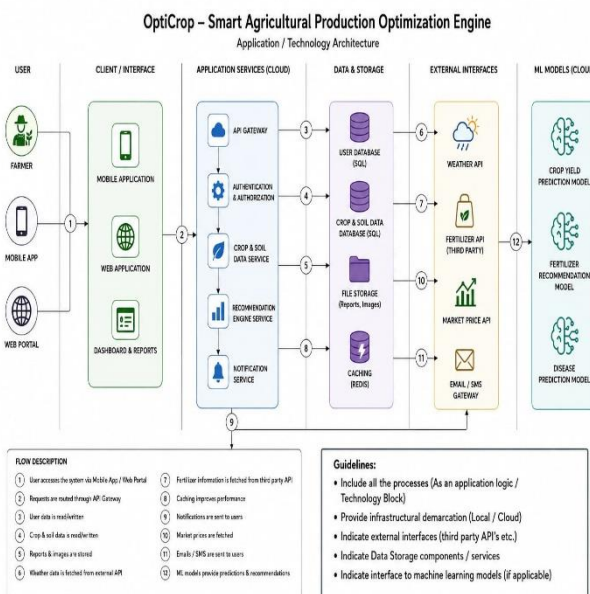
Project Design Phase-II

Data Flow Diagram & User Stories

Date	27 June 2026
Project Name	OptiCrop
Maximum Marks	4 Marks

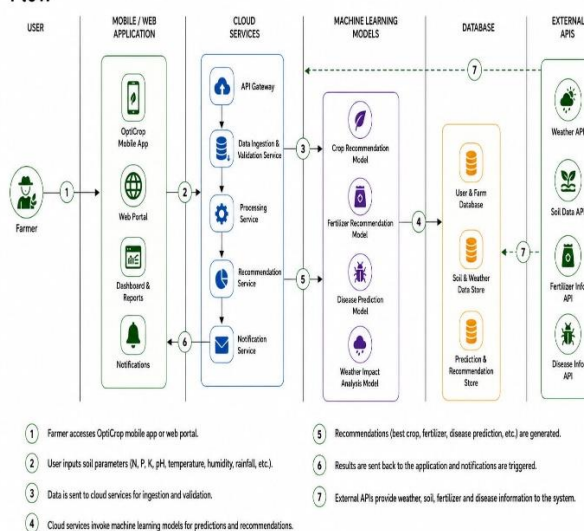
Data Flow Diagrams:

The **Data Flow Diagram (DFD)** for the **OptiCrop – Smart Agricultural Production Optimization Engine** illustrates how data moves through the system. It shows the flow of soil nutrient values (Nitrogen, Phosphorus, Potassium), pH, temperature, humidity, and rainfall data from the user to the application. The system validates and processes the input, applies the machine learning model to predict the most suitable crop, and displays the recommendation to the user. The DFD also represents how datasets and trained models are used during the prediction process.



OptiCrop – Smart Agricultural Production Optimization Engine

Flow



User Stories

User Type	Functional Requirement (Epic)	User Story	User Story / Task	Acceptance criteria	Priority	Release
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		Number				
Customer (Farmer)	Enter Crop Parameters	USN-1	As a farmer, I can enter soil nutrient values (N, P, K), pH, temperature, humidity, and rainfall.	The system accepts valid input values.	High	Sprint-1
Customer (Farmer)	Input Validation	USN-2	As a farmer, I want the system to validate my inputs before prediction.	Invalid or missing values display an error message.	High	Sprint-1
Customer (Farmer)	Crop Recommendation	USN-3	As a farmer, I want to receive the most suitable crop recommendation based on my input values.	The system predicts and displays the recommended crop accurately.	High	Sprint-1
Customer (Farmer)	View Results	USN-4	As a farmer, I want to view the crop recommendation in a clear and understandable format.	The recommendation is displayed successfully on the screen.	Medium	Sprint-1
Customer (Farmer)	Prediction History (Optional)	USN-5	As a farmer, I want to view my previous crop predictions.	The system displays saved prediction records (if implemented).	Low	Sprint-2
Administrator	Dataset Management	USN-6	As an administrator, I want to	The updated dataset is processed	Medium	Sprint-2

			update the crop dataset and retrain the machine learning model when required.	successfully, and the model is retrained.		
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